

Vitamins

Vitamins are organic substances (i.e., containing carbon atoms) needed in very small amounts to perform specific metabolic functions (71, 166). Vitamins typically act as coenzymes, facilitating numerous reactions in the body. For instance, B vitamins help the body make energy from the metabolism of carbohydrates. [Table 10.13](#) describes the functions and some food sources of individual vitamins.

TABLE 10.13 Vitamins

Vitamin	Function	Food sources (Oregon State University)	DRI and UL ^b per day
Vitamin A	Necessary for vision, healthy skin, teeth, body tissues, and healthy mucous membranes and skin	Animal foods including animal liver (veal, beef, goose, lamb, turkey), meat, fortified milk, cheese, herring	Males: 300-900 mcg RAE (retinal activity equivalents) (RDA) Females: 300-1,300 mcg RAE (RDA) Males: 600-3,000 mcg RAE (UL) Females: 600-3,000 mcg RAE (UL)
Beta carotene	An antioxidant ^a ; converted into vitamin A in the body	Sweet potato, carrots, pumpkin, spinach, collards, kale, winter squash, lamb quarters, beet greens, turnip greens, cabbage	No RDA or AI; however, 1 IU beta carotene from food = 0.05 mcg RAE vitamin A UL is not determinable due to lack of data. Supplementation is advised only for individuals at risk for vitamin A deficiency.
Vitamin D	Aids calcium absorption, helps maintain blood levels of calcium and	Fish (swordfish, salmon, tuna, sardines, mackerel, carp, eel, whitefish), fortified milk,	Males: 15-20 mcg (600-800 IU) (RDA) Females: 15-20 mcg (600-800 IU) (RDA)

Vitamin	Function	Food sources (Oregon State University)	DRI and UL^b per day
	phosphorus; necessary for building bone mass and preventing bone loss	fortified breakfast cereals, egg yolks	Males: 63-100 mcg (2,500-4,000 IU) (UL) Females: 63-100 mcg (2,500-4,000 IU) (UL)
Vitamin E	An antioxidant ^a ; needed for immune functioning and metabolism	Oils (wheat germ, vegetable), fortified breakfast cereals, nuts and seeds, wheat germ, peanut butter, corn oil	Males: 6-15 mg (7.5-22.4 IU) (RDA) Females: 6-19 mg (9-28.4 IU) (RDA) Males: 200-1,000 mg (300-1,500 IU) (UL) Females: 200-1,000 mg (300-1,500 IU) (UL)
Vitamin K (phylloquinone)	Needed for blood clotting; supports tissue and bone health	Dark green leafy vegetables (kale, Brussels sprouts, spinach, chard, turnip and mustard greens, beet greens, radicchio), broccoli, asparagus, lamb quarters	Males: 30-120 mcg (AI) Females: 30-90 mcg (AI) No UL due to lack of data
Vitamin C	Promotes healthy cell development, wound healing, and resistance to infections. Serves as an antioxidant ^a ; necessary for conversion of the inactive form of folic acid to the active form; makes iron available for hemoglobin synthesis.	Sweet peppers, peaches, guava, broccoli, kiwifruit, citrus fruit (strawberries, oranges, limes, lemons, grapefruit, tangerine), papayas, cantaloupe, tomatoes, potatoes, onions	Males: 15-90 mg (RDA) Females: 15-120 mg (RDA) Males: 400-2,000 mg (UL) Females: 400-2,000 mg (UL)
Thiamin (B ₁)	Coenzyme for carbohydrate metabolism; needed for normal functioning	Fortified breakfast cereals, sunflower seeds, peas, pork, oranges, orange juice,	Males: 0.5-1.2 mg (RDA) Females: 0.5-1.4 mg (RDA)

Vitamin	Function	Food sources (Oregon State University)	DRI and UL^b per day
	of the nervous system and muscles, including the heart	lima beans, pecans, enriched rice	No UL due to lack of data
Riboflavin (B ₂)	Coenzyme in red blood cell formation; nervous system functioning; and metabolism of carbohydrate, protein, and fat; needed for vision and may help protect against cataracts	Liver, wheat germ, brewer's yeast, almonds, cheese, fortified breakfast cereal, whey protein, milk, eggs, lamb, pork, veal, beef, broccoli, yogurt	Males: 0.5-1.3 mg (RDA) Females: 0.5-1.6 mg (RDA) No UL due to lack of data
Niacin	Coenzyme for carbohydrate, protein, and fat metabolism and proper nervous system functioning; high intakes can lower elevated cholesterol	Soy protein, soy flour, textured vegetable protein, whey protein, beef, peanuts, peanut butter, sunflower seeds, fortified breakfast cereals	Males: 6-16 mg (RDA) Females: 6-18 mg (RDA) Males: 10-35 mg (UL) Females: 10-35 mg (UL)
Pyridoxine (B ₆)	Coenzyme for protein metabolism and nervous and immune system function; involved in synthesis of hormones and red blood cells	Liver, bananas, fortified breakfast cereals, soybeans, chicken, tuna, raw carrots, beef, broccoli, spinach, potatoes, alfalfa sprouts, navy beans, peanut butter, garbanzo beans, walnuts, sunflower seeds, avocados, eggs, lima beans, cabbage, salmon	Males: 0.5-1.7 mg (RDA) Females: 0.5-2.0 mg (RDA) Males: 30-100 mg (UL) Females: 30-100 mg (UL)
Folate	Needed for normal growth and development and red blood cell formation; reduces risk of neural tube birth defects; may reduce risk of heart	Brewer's yeast, fortified breakfast cereals, liver, black-eyed peas, beans (pinto, black, lima, white, garbanzo, soy), peanuts, peanut butter, spinach, turnip greens,	Males: 150-400 mcg (RDA) Females: 150-600 mcg (RDA) Males: 300-1,000 mcg (UL)

Vitamin	Function	Food sources (Oregon State University)	DRI and UL^b per day
	disease and cervical dysplasia	asparagus, mustard greens, seaweed, eggs, enriched bread, oranges, orange juice	Females: 300-1,000 mcg (UL)
Cobalamin (B ₁₂)	Vital for blood formation and healthy nervous system	Liver, oysters, lamb, eggs, beef, shellfish, fish, poultry, pork, chicken, fortified breakfast cereals	Males: 0.9-2.4 mcg (RDA) Females: 0.9-2.8 mcg (RDA) No UL due to lack of data
Biotin	Assists in the metabolism of fatty acids and use of B vitamins	Nuts (peanuts, hazelnuts, almonds, cashews, macadamia), soybeans, peanut butter, black-eyed peas, liver, milk, egg yolks, yeast, cheese, cauliflower, carrots, avocados, sweet potatoes	Males: 8-30 mcg (AI) Females: 8-35 mcg (AI) No UL due to lack of data
Pantothenic acid	Aids in normal growth and development	Liver, sunflower seeds, fortified breakfast cereals, egg yolks, whey protein, soy protein, peanuts, peanut butter, pecans, veal, enriched rice, broccoli, lima beans	Males: 2-5 mg (AI) Females: 2-7 mg (AI) No UL due to lack of data

Vitamin	Function	Food sources (Oregon State University)	DRI and UL^b per day
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^aAntioxidants are substances that modify cell-signaling pathways and counteract oxidative stress-induced cell damage caused by reactive oxygen and nitrogen species (free radicals). Free radicals are both beneficial and harmful to human health. An overproduction of, or overexposure to, free radicals combined with a deficiency in antioxidants can result in oxidative stress and subsequent damage to cellular lipids, proteins, and DNA (71, 166).

^bDietary Reference Intake (DRI): A set of nutrient recommendations that include the values Recommended Dietary Allowance (RDA), Adequate Intake (AI), Estimated Average Requirement (EAR), and the Tolerable Upper Level Intake (UL). Tolerable Upper Level Intake (UL): The highest daily intake of a nutrient that is likely to not pose a risk of adverse health effects in most of the general population. DRIs and UL are listed as a range across the life span for each sex. DRI and UL values from (71-74).

Data from National Institutes of Health.

Water-soluble vitamins, including the B vitamins and vitamin C, dissolve in water and are transported in the blood. With the exception of vitamin B₁₂, which is stored in the liver for years, water-soluble vitamins are not stored in appreciable amounts in the body; instead the body uses what is needed and then excretes the remaining amounts in urine (165). Though there are no known side effects from excess vitamin B₁₂ consumption, consuming more than the body can use will not increase energy or improve health (18, 74). Vitamins A, D, E, and K are fat soluble and therefore carried by fat in the blood and stored in fat tissue in the body (109). Excess preformed vitamin A (not beta carotene, alpha carotene, or beta cryptoxanthin, which are converted into vitamin A in the body) is toxic and associated with significant adverse effects including liver damage, intracranial pressure (pseudotumor cerebri), dizziness, nausea, headaches, skin irritation, pain in joints and bones, coma, and even death (23). Excess vitamin A intake can occur through diet but typically results from high levels of vitamin A in supplements (72, 109). Toxic levels of vitamin D can lead to heart arrhythmias and increased levels of blood calcium, which can cause blood vessel and tissue calcification, as well as damage to the heart, blood vessels, and kidneys (109). Vitamin E acts as an anticoagulant and therefore

thins the blood. Regular intake of excessive amounts of vitamin E can lead to high serum vitamin E levels, which is associated with increased risk of hemorrhagic stroke, particularly in individuals on blood thinners (125). Because vitamin K helps blood clot, excess intake of this vitamin can interfere with the effects of some anticlotting medications such as warfarin (Coumadin) (94).

Vitamin D is a particular vitamin of interest for both sedentary and athletic individuals because it is important for immunity, protein synthesis, muscle function, cardiovascular function, inflammatory response, cell growth, and regulation of muscle and bone (39, 121). Many people, including athletes, struggle to meet the recommendations for vitamin D due to few foods being rich sources of this vitamin (51). According to the Scientific Report of the 2020 Dietary Guidelines Advisory Committee, more than 90 percent of men and women over 19 years of age do not consume enough vitamin D (56, 154). Eating a diet containing seafood, and choosing foods that are fortified with vitamin D, including milk, fortified soy beverages, fortified soy yogurt, and some whole-grain cereals, can help adults meet their needs for vitamin D. In addition to dietary sources, the body can make vitamin D from the sun through sun exposure on the skin; however, populations living at higher latitudes or who spend a lot of time indoors do not have enough sun exposure to meet vitamin D needs (51). As such, vitamin D supplementation may be necessary for many individuals to prevent deficiency. A meta-analysis of 11 randomized control trials and 436 athletes showed that vitamin D supplementation only had a small or trivial effect on upper and lower body muscular strength or power (150). Despite lack of a notable performance effect, adequate vitamin D levels may be important for athletes to protect against stress fractures and acute illness (35, 39, 91).

Minerals

Minerals contribute to the structure of bone, teeth, and nails; are a component of enzymes; and perform a wide variety of metabolic functions. For example, calcium is needed for bone and tooth formation and function, nerve transmission, and muscle contraction. Iron is necessary for oxygen transport and is also a component of enzymes necessary for energy metabolism. Calcium; phosphorus; magnesium; iron; and the **electrolytes** sodium, potassium, and chloride are often called the *major minerals*. For the athlete, minerals are important for bone health, oxygen-carrying capacity, and fluid and electrolyte balance. The minerals, their functions, and good food sources are listed in [table 10.14](#).

TABLE 10.14 Minerals

Mineral	Function	Food sources	DRI and UL* per day
Calcium	Essential for developing and maintaining healthy bones and teeth; assists with blood clotting, muscle contraction, and nerve transmission	Fruit juices and fruit drinks fortified with calcium, cheese, sardines, milk, cottage cheese, yogurt, ice cream, calcium-set tofu, turnip greens, Chinese cabbage, mustard greens, kale, rutabaga	Males: 700-1,300 mg (RDA) Females: 700-1,300 mg (RDA) Males: 2,000-3,000 mg (UL) Females: 2,000-

*Dietary Reference Intake (DRI): A set of nutrient recommendations that include the values Recommended Dietary Allowance (RDA), Adequate Intake (AI), Estimated Average Requirement (EAR), and the Tolerable Upper Level Intake (UL). Tolerable Upper Level Intake (UL): The highest daily intake of a nutrient that is likely to not pose a risk of adverse health effects in most of the general population. DRIs and UL are listed as a range across the life span for each sex. DRI and UL values from (72, 73).

Data from National Institutes of Health.

Mineral	Function	Food sources	DRI and UL* per day
			3,000 mg (UL)
Phosphorus	Works with calcium to develop and maintain strong bones and teeth; enhances use of other nutrients; essential for energy metabolism, DNA structure, and cell membranes	Cheese, fish, beef, pork, whole-wheat products, cocoa powder, pumpkin seeds, sunflower seeds, almonds	Males: 460-1,250 mg (RDA) Females: 460-1,250 mg (RDA) Males: 3-4 g (UL) Females: 3-4 g (UL)
Magnesium	Activates nearly 100 enzymes and helps nerves and muscles function; constituent of bones and teeth	Bran (wheat and rice), cocoa powder, fortified breakfast cereals, seeds (pumpkin, sunflower), soybeans, nuts (almonds, pine nuts, hazelnuts, cashews, walnuts, peanuts), spinach	Males: 80-420 mg (RDA) Females: 80-400 mg (RDA) Males: 65-350 mg (UL) Females: 65-350 mg (UL)
Molybdenum	Needed for metabolism of DNA and ribonucleic acid (RNA) and production of uric acid	Milk, milk products, peas, beans, liver, whole-grain products	Males: 17-45 mcg (RDA) Females: 17-50 mcg (RDA) Males: 300-2,000

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Data from National Institutes of Health.

Mineral	Function	Food sources	DRI and UL* per day
			mcg (UL) Females: 300-2,000 mcg (UL)
Manganese	Necessary for the normal development of the skeletal and connective tissues; involved in metabolism of carbohydrates	Wheat germ, wheat bran, rice bran, fortified breakfast cereals, rice cakes, nuts (peanuts, pecans, pine nuts, walnuts, almonds, hazelnuts), soybeans, mussels, whole-wheat products (pastas, breads, and crackers)	Males: 1.2-2.3 mg (AI) Females: 1.2-2.6 mg (AI) Males: 2-11 mg (UL) Females: 2-11 mg (UL)
Copper	Involved in iron metabolism, nervous system functioning, bone health, and synthesis of proteins; plays a role in the pigmentation of skin, hair, and eyes	Liver, shellfish (especially oysters), lobster, nuts (cashews, Brazil nuts, hazelnuts, walnuts, peanuts, almonds, pecans, pistachios), seeds (sunflower, pumpkin), fortified breakfast cereals, great northern beans	Males: 340-900 mcg (RDA) Females: 340-1,300 mcg (RDA) Males: 1,000-10,000 mcg (UL) Females: 1,000-10,000 mcg (UL)

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Data from National Institutes of Health.

Mineral	Function	Food sources	DRI and UL* per day
Chromium	Aids in glucose metabolism and may help regulate blood sugar and insulin levels in people with diabetes	Mushrooms (white), raw oysters, wine, apples, brewer's yeast, beer, pork, chicken	Males: 11-35 mcg (AI) Females: 11-45 mcg (AI) No UL due to lack of data
Iodine	Part of the thyroid hormone; helps regulate growth, development, and energy metabolism	Iodized salt, saltwater fish and seafood	Males: 90-150 mcg (RDA) Females: 90-290 mcg (RDA) Males: 200-1,100 mcg (UL) Females: 200-1,100 mcg (UL)
Iron	Necessary for red blood cell formation and function; constituent of myoglobin and component of enzyme systems	Liver, beef, lamb, pork, veal, poultry, clams, oysters, fortified breakfast cereals, enriched bread products, brewer's yeast, nuts (pine nuts, cashews, almonds), beans (kidney, green, garbanzo)	Males: 7-11 mg (RDA) Females: 7-27 mg (RDA) Males: 40-45 mg (UL) Females: 40-45 mg (UL)

*Dietary Reference Intake (DRI): A set of nutrient recommendations that include the values Recommended Dietary Allowance (RDA), Adequate Intake (AI), Estimated Average Requirement (EAR), and the Tolerable Upper Level Intake (UL). Tolerable Upper Level Intake (UL): The highest daily intake of a nutrient that is likely to not pose a risk of adverse health effects in most of the general population. DRIs and UL are listed as a range across the life span for each sex. DRI and UL values from (72, 73).

Data from National Institutes of Health.

Mineral	Function	Food sources	DRI and UL* per day
Selenium	Essential component of a key antioxidant enzyme; necessary for normal growth and development and for use of iodine in thyroid function	Tenderloin of beef, pollock, trout, tuna, oysters, mackerel, flounder, liver, sunflower seeds, wheat bran, wheat germ, some pork, fortified breakfast cereals, perch, crab, clams, cod, haddock, whole-wheat breads	Males: 20-55 mcg (RDA) Females: 20-70 mcg (RDA) Males: 90-400 mcg (UL) Females: 90-400 mcg (UL)
Zinc	Essential part of more than 100 enzymes involved in digestion, metabolism, reproduction, and wound healing	Oysters, beef, veal, lamb, pork, chicken, lima beans, black-eyed peas, white beans	Males: 3-11 mg (RDA) Females: 3-13 mg (RDA) Males: 7-40 mg (UL) Females: 7-40 mg (UL)

*Dietary Reference Intake (DRI): A set of nutrient recommendations that include the values Recommended Dietary Allowance (RDA), Adequate Intake (AI), Estimated Average Requirement (EAR), and the Tolerable Upper Level Intake (UL). Tolerable Upper Level Intake (UL): The highest daily intake of a nutrient that is likely to not pose a risk of adverse health effects in most of the general population. DRIs and UL are listed as a range across the life span for each sex. DRI and UL values from (72, 73).

Data from National Institutes of Health.

Two minerals, iron and calcium, deserve additional attention. Athletes who do not consume enough dietary iron can develop iron deficiency or iron deficiency **anemia**, both of which can impair performance. Inadequate dietary calcium can contribute to low bone density and possibly future risk of developing osteopenia or osteoporosis.

Iron

Iron is essential for both the functioning and synthesis of hemoglobin, a protein that transfers oxygen throughout the body (158). In addition, iron is a component of the protein myoglobin, which transports oxygen to muscles (63). This mineral plays important roles in growth, development, cell functioning, and the synthesis and functioning of some hormones (30, 160).

Iron deficiency and iron deficiency anemia is a major public health concern worldwide, mainly affecting women and children (40). Though it disproportionately affects those in developing countries, it is common in industrialized countries as well. The National Health and Nutrition Examination Survey (NHANES) from 2015-16 found that approximately 11% of American females aged 12 to 49 had iron deficiency, with the goal to reduce that level to 7.2% (99). (In studies examining iron deficiency in female athletes, more than 25% tested positive for iron deficiency [65, 102].) Iron deficiency occurs in three stages (in order of severity): depletion, marginal deficiency, and anemia (90, 117). Iron carries oxygen to the working muscles, and even marginal iron deficiency may impair athletic performance (19, 63). Iron deficiency anemia develops when low iron stores persist for a period of time and the body cannot make enough healthy red blood cells to deliver oxygen throughout the body (158). Symptoms depend on the individual; some people are asymptomatic or get used to their symptoms and assume that they are normal. Symptoms of iron deficiency or iron deficiency anemia may include weakness, fatigue, irritability, poor concentration, headache, decreased exercise capacity, hair loss, and dry mouth (15). Other symptoms associated with iron deficiency anemia include feeling cold often, inflamed tongue (glossitis), shortness of breath during routine activities, and pica (the desire to eat nonfood substances such as laundry starch, dirt, clay, and ice) (52). Iron deficiency has also been shown to negatively affect mental health and contribute to depression (30, 99).

Women of childbearing age, teenage girls, pregnant women, infants, and toddlers have the greatest need for iron and therefore an increased risk of becoming deficient. In addition, distance runners, vegetarian athletes, female athletes, those who lose a significant amount of blood during their menstrual cycle, people who take excessive amounts of antacids, and people with certain digestive diseases such as celiac disease have an increased risk of developing iron deficiency anemia (72).

The two types of iron found in food are heme and nonheme iron. Heme iron, derived from hemoglobin, is found in foods that originally contained hemoglobin and myoglobin—animal foods including red meats, fish, and poultry. Heme iron is absorbed better than nonheme iron, and absorption is not affected by anything else we eat. Approximately 15% to 35% of the heme iron consumed is actually absorbed (112).

Nonheme iron is the form of iron found in all other, non-meat foods including vegetables, grains, and iron-fortified breakfast cereal. Only 2% to 20% of nonheme iron is absorbed (160). While heme iron is not affected by compounds found in food eaten concurrently, many factors affect the absorption of nonheme iron. For instance, the nonheme iron in spinach is bound to a substance called phytic acid, the storage form of phosphorus in plants. Phytic acid decreases the absorption of nonheme iron. In addition, several other substances may decrease the absorption of nonheme iron, including tannins (found in tea and wine), calcium (found in dairy and multivitamins), polyphenols, phytates (found in legumes and whole grains), and some of the proteins in soy. A person can increase the amount of nonheme iron absorbed by consuming vitamin C-rich foods or beverages at the same time or consuming a nonheme source at the same time heme iron is consumed. For example, pairing spinach with meat increases iron absorption from spinach (72, 160).

There are many supplemental forms of iron; each supplies a different amount of elemental iron while also varying in

bioavailability and the potential for side effects such as stomach upset (103, 157). Ferrous iron has a high bioavailability and is most recommended; it is often paired with vitamin C, which enhances iron absorption (157). It is recommended to take iron supplements on an empty stomach, at least an hour before meals, because many foods, drinks, and other dietary supplements can interfere with iron absorption. For example, whole grains, coffee, black tea, red wine, and hot chocolate can impair iron absorption. Additionally, both calcium and supplemental magnesium may interfere with iron absorption (169). Because of the many factors affecting iron intake and absorption, including intake of other minerals the athlete may not be consuming in adequate quantities, only a medical doctor or registered dietitian should recommend iron supplements and how to take an iron supplement for maximum absorption and minimal stomach upset (157).

Calcium

Adequate calcium intake throughout childhood and adolescence is essential for the development of strong bones. Calcium helps bones grow in length and density during adolescence, with up to 90% of peak bone mineral density occurring during late adolescence (61). In adults, calcium helps maintain bone density. When dietary calcium intake falls short, calcium is pulled from its storage site in bone to meet the demands of the body and keep calcium concentrations in the blood, muscle, and intercellular fluids constant. Calcium is essential for attaining peak bone mass, and calcium deficiencies can impair the attainment of peak bone mineral density and increase risk of fracture later in life (109). Calcium also keeps teeth strong, helps regulate muscle contraction, and plays a role in nerve functioning, blood vessel expansion and contraction, and hormone and enzyme secretion (109).

The NHANES found that just 15% of 9- to 13-year-old females and less than 10% of females aged 14 to 18 and over the age of 51 met the adequate intake (AI) for calcium from diet alone (11).

Therefore, athletes should be encouraged to include dairy products and other calcium-rich foods in their diets. The athlete's medical doctor or registered dietitian may suggest a calcium supplement if calcium needs cannot be met through diet alone.

Caloric Versus Nutrient-Dense Foods

Given the rise in obesity in America, some classify the American diet as calorie dense but nutrient poor. The Scientific Report of the 2020 Dietary Guidelines Advisory Committee recommends a dietary pattern including a variety of nutrient-dense foods. Though there is no standardized definition of the term **nutrient density**, in general, choosing nutrient-dense foods means looking for foods based on the nutrients such as vitamins, minerals, and fiber, as well as the healthy plant-based compounds that they provide, whereas *caloric density* refers to the caloric content of a food. Foods high in nutrient density include milk, vegetables, protein foods, and grains, while foods high in caloric density but low in nutrient density typically include potato chips, desserts, and candy (44). For more information on nutrient versus caloric density, the reader is referred to the work of Drewnowski (44).

Fluid and Electrolytes

Water is the largest component of the body, representing 45% to 75% of a person's body weight (75). In the human body, water acts as a lubricant, shock absorber, building material, and solvent. Additionally, water is essential for body temperature regulation (water loss through sweat helps cool off skin, particularly in hot environments and during exercise), nutrient transport and waste product removal, and maintaining fluid balance and therefore normal blood pressure (81). Water is so important that even under optimal temperature conditions, the body can survive only a few days without water.

While maintaining adequate **hydration** is important for all individuals, athletes must pay close attention to their hydration status since sweat losses that exceed fluid intake can quickly lead to a hypohydrated state with a subsequent increase in core body temperature, decrease in blood plasma volume, and increase in heart rate and perceived exertion (32, 139). When this happens, sweat output cannot keep up with increases in core body temperature unless fluids are given. Repeated exercise in hot environments helps the body adapt to heat stress (e.g., greater sweat volume, lower electrolyte concentration of sweat, and lower temperature for the onset of sweating), and therefore athletes may be more prone to **dehydration** and heat stress at the beginning of the season (54). In addition, athletes with less training may be more prone to heat stress than trained athletes (32). Due to physiological changes that alter water conservation in the body as well as age-related decline in fluid intake, the elderly have an increased risk of dehydration and **hypohydration** (32). Children may have a greater risk of dehydration as well, resulting from increased heat gain from the environment due to greater surface area–body mass ratio compared to adults, increased heat production during exercise, decreased ability to dissipate heat through sweat, and a decreased sensation of thirst compared to adults (12, 46). Additionally, those

with sickle cell trait, cystic fibrosis, and some other diseases have an increased risk of becoming dehydrated (12, 163). Even mild dehydration, representing 2% to 3% weight loss, can increase core body temperature and significantly affect athletic performance by increasing fatigue and decreasing motivation, neuromuscular control, accuracy, power, strength, muscular endurance, and overall performance (13, 26, 42, 58, 84, 85, 141, 153). Dehydration can increase core body temperature, reduce stroke volume and cardiac output, decrease blood pressure, reduce blood flow to muscles, increase heartbeat, exacerbate symptomatic exertional rhabdomyolysis, and increase risk of heatstroke and death (3, 32, 108). One's risk for dehydration, and subsequent heat stress, is greater in hot, humid environments and at altitude (27, 108, 146).

Fluid Balance

Fluid balance is the balance between water intake from food, drinks, and metabolic water and water output through expired air, insensible perspiration, sweat, feces, and urine. It is important for a person to consume adequate sources of water to prevent dehydration from excess water loss. The AI for water is 3.7 liters (125.1 fl oz or 15.6 cups) and 2.7 L (91.3 fl oz or 11.4 cups) per day, respectively, for men and women. However, the AI for pregnant and lactating women is 3 liters (101.4 fl oz or 12.7 cups) and 3.8 liters (128.5 fl oz or 16.1 cups) per day, respectively. All sources of fluid, including beverages such as coffee, tea, juice, and soda, as well as the fluid in foods, contribute to meeting a person's water needs from dietary sources (75).

Maintaining fluid balance during training and competition can be a challenge for many athletes, particularly those who sweat profusely or train in hot, humid environments or at altitude. In addition to environmental conditions, clothing, equipment, and larger body size can increase sweat rate, while unsafe weight loss practices such as frequent use or overuse of diuretics or laxatives can increase risk of dehydration (32, 38). American football players, particularly linemen,

have a greater risk of becoming dehydrated because of their equipment and, in many cases, larger body size (32). For example, a study in National Football League players found that backs and receivers, with an average body mass of 93 ± 6 kilograms (205.0 ± 13.2 lb), lost an average of 1.4 ± 0.45 liters (47.3 ± 15.2 fl oz; or approximately 6 ± 2 cups) of sweat per hour, while linemen, with an average body mass of 135.6 ± 17 kilograms (298.9 ± 37.5 lb), lost an average 2.25 ± 0.68 liters (76.1 ± 23.0 fl oz; approximately 9.5 ± 2.9 cups) of sweat per hour during practice. Calculated sweat losses for both groups during a total of 4.5 hours of practice on days with two practice sessions were 6.4 ± 2.0 liters (216.4 ± 67.6 fl oz; approximately 27.1 ± 8.5 cups) for backs and receivers and 10.1 ± 3.1 liters (341.5 ± 104.8 fl oz; 42.7 ± 13.1 cups) for linemen (55). Multiple layers of clothing and protective equipment contribute to sweat losses and dehydration risk in hockey players (14), while intentional dehydration and other unsafe weight loss practices can increase dehydration risk in wrestlers (170). In addition, it is important to note that a very wide range of fluid losses through sweat exists. For instance, National Basketball Association players lost from 1.0 to 4.6 liters (33.8-155.5 fl oz; 4.2-19.4 cups) and had a mean loss of 2.2 ± 0.8 liters (74.4 ± 27.1 fl oz; 9.3 ± 3.4 cups) of sweat over the course of a 40-minute game in which the average playing time was 21 ± 8 minutes (120). To reduce the risk of dehydration and heat stress, athletes should avoid exercising under extremely hot, humid conditions and should adequately replace fluid losses from prolonged exertion.

Preventing Dehydration

Given the negative effects associated with dehydration, athletes should try to prevent water weight losses exceeding 2% of body weight while also restoring electrolytes lost through sweat (3, 139). The first step to preventing dehydration is to assess hydration status ([table 10.15](#)). Urine specific gravity (USG) can be used to assess hydration status because the test is easy to use, inexpensive, and

portable (168). However, USG is not a sensitive indicator of acute changes in hydration; instead it is better as a measure of chronic hydration status (120, 131). A quick and simple method for estimating hydration status involves measuring changes in body weight from before to after a workout. Athletes should weigh themselves in minimal, lightweight clothing, after drying off and urinating, immediately before and after their workout. Sweaty clothes should be removed before weighing. Each pound (0.45 kg) lost during practice represents 16 ounces (0.47 L) of fluid. A loss of 2% or more of body weight indicates the athlete is not adequately replacing fluid lost through sweat (3). In addition to identifying acute dehydration from one workout, assessing weight changes over time may help identify athletes who are chronically dehydrated—those who lose several pounds over the course of a few days (24). See [table 10.15](#) for biomarkers of proper or sufficient hydration.

TABLE 10.15 Biomarkers of Hydration Status

Measure	Practicality	Validity (acute vs. chronic changes)	EUH cutoff
Total body water	Low	Acute and chronic	<2%
Plasma osmolality	Medium	Acute and chronic	<290 mOsm
Urine specific gravity	High	Chronic	<1.020 g/mL
Urine osmolality	High	Chronic	<700 mOsm
Body weight	High	Acute and chronic*	<1%

Note: EUH = euhydration.

*Potentially confounded by changes in body composition during very prolonged assessment periods.

Reprinted by permission from American College of Sports Medicine, M.N. Sawka, L.M. Burke, et al., "American College of Sports Medicine Position Stand: Exercise and Fluid Replacement," *Medicine & Science in Sports & Exercise* 39, no. 2 (2007): 377-390.

In addition to identifying athletes who do not adequately hydrate, one can calculate sweat rate, thereby giving a better idea of fluid needs during exercise, by weighing athletes before exercise and

again after an intense 1-hour practice session while also measuring fluid intake and urine volume produced. Sweat rate is equal to preexercise body weight minus postexercise body weight + fluid intake during exercise – urine produced (14, 24). Assessments of urine quantity, through USG or urine volume, may be misleading during the immediate rehydration period after dehydration. When athletes consume large quantities of hypotonic fluid, they produce copious amounts of urine long before they become adequately hydrated (147, 148).

In addition to monitoring weight changes, athletes are sometimes advised to check their urine color. However, the relationship between urine color and hydration status is very subjective (3, 108). Also, beets, blackberries, certain food colors, and medications can turn urine pink, red, or light brown (9). In addition, B vitamins, carotenoids (such as beta carotene), and some medications can turn urine dark yellow, bright yellow, or orange, while artificial food colors (such as those found in some sport drinks) may also turn urine blue or green (9).

A very wide range of fluid losses, in the form of sweat, exists among athletes. Therefore each athlete should develop an individualized hydration plan.

Electrolytes

The major electrolytes lost in sweat include sodium chloride and, to a lesser extent and in order, potassium, magnesium, and calcium (89). Sodium influences fluid regulation by helping retain more of the fluid consumed (107). In addition, all of the electrolytes lost through sweat are essential to muscle contraction and nerve conduction. Thus, any disturbance in the balance of electrolytes in body fluids could potentially interfere with performance. Sodium losses through sweat vary tremendously between athletes, with reported concentrations ranging from 0.2 to greater than 12.5 g/L (10 to over 544 mEq/L) (3, 32). Given the large quantity of sodium

some athletes lose through sweat, replacing sodium losses may require a conscious decision to choose higher-sodium foods, salt their food, and add electrolytes to their sport drinks. Athletes who exercise intensely or for hours and hydrate excessively with only water or a no- or low-sodium beverage may dilute their blood sodium levels to dangerously low levels—below 130 mmol/L—a condition called **hyponatremia**. This leads to intracellular swelling, and, when blood sodium levels fall below 125 mmol/L, to headaches, nausea, vomiting, muscle cramps, swollen hands and feet, restlessness, and disorientation. When blood sodium drops below 120 mmol/L, risk of developing cerebral edema, seizures, coma, brain stem herniation, respiratory arrest, and risk of death increase (2, 3, 6). To avoid hyponatremia, fluid intake should not exceed sweat losses (athletes should not weigh more after they finish exercising than they did at the start of their training session), and athletes should consume sodium through either sport drinks or food (3, 120).

Sport drinks provide small amounts of potassium to replace sweat potassium losses. However, they are not a significant contributor to total potassium intake. Because research shows that less than 2% of adults in the United States meet the dietary recommendations for potassium (31), athletes should focus on consuming more potassium-rich foods in their diet, such as tomatoes, citrus fruits, melons, potatoes, bananas, and milk.

Athletes who exercise intensely or for hours and hydrate excessively with only water or a no- or low-sodium beverage may dilute their blood sodium to dangerously low levels.

Fluid Intake Guidelines

Ideally, athletes should start exercise or training in a hydrated state, avoid losing more than 2% of body weight (due to sweat losses) during exercise, and rehydrate completely after exercise and before

the next training session. The amount of fluid necessary to fully achieve rehydration depends on the time period before the next bout of training. However, studies show that some athletes start practice or competition in a hypohydrated state, making it more difficult to consume enough fluid during the exercise session to make up for poor preexercise hydration status (120, 134). Additionally, athletes with significant sweat losses may not voluntarily drink enough fluid to adequately rehydrate and prevent dehydration during training and competition. A systematic approach to fluid replacement is necessary for this reason and also because thirst may not be a reliable indicator of fluid needs for athletes who are heavy sweaters or those who are practicing intensely in hot environmental conditions (24). Though fluid and electrolyte guidelines should be individualized as much as possible, given the wide range of fluid and electrolyte losses in athletes, the general guidelines presented next are a good starting place with athletes until specific recommendations can be provided based on sweat rate (6). Additionally, strength and conditioning professionals should ensure that athletes are given adequate time to drink and access to cool fluids (10-15 °C [50-59 °F]) (82).

Before Activity

Prehydrate, if necessary, several hours before exercise to allow for fluid absorption and urine output (3).

During Activity

Due to large variations in sweat rates and electrolyte concentrations, athletes should measure weight changes during training and competition in specific weather conditions and develop individualized hydration strategies based on this information (3). During prolonged activity in hot weather, the IOM recommends sport drinks containing 20 to 30 mEq of sodium (460-690 mg with chloride as the anion) per liter, 2 to 5 mEq of potassium (78-195 mg) per liter, and carbohydrate in a concentration of 5% to 10% (70). Additionally,

when a sport drink is ingested at high rates during intense or prolonged exercise, athletes may want to choose one that contains multiple types of carbohydrates with different intestinal transport mechanisms, such as glucose, fructose, and maltodextrin. Ingestion of multiple types of carbohydrate versus a single carbohydrate will lead to greater gastric emptying, carbohydrate absorption, oxidation, and possibly better performance (36, 83). All beverages provided should be cool—10-15 °C (50-59 °F)—but not cold (24).

The American Academy of Pediatrics recommends periodic drinking be enforced in children. These guidelines suggest that children weighing 88 pounds (40 kg) drink 5 ounces (148 mL) of cold water or a flavored salted beverage every 20 minutes during practice and that adolescents weighing 132 pounds (60 kg) drink 9 ounces (266 mL) even if they do not feel thirsty. Another recommendation is a sodium chloride concentration of 15 to 20 mmol/L (1 g per 2 pints), which has been shown to increase voluntary hydration by 90% when compared to unflavored water (34, 171).

After Activity

After exercise, athletes should replace fluid and electrolyte losses. If time permits, normal meals, snacks (provided that they contain some sodium), and water will restore fluid and electrolyte losses. Additional salt can be added to foods when sweat sodium losses are substantial (3, 75). If dehydration is significant, or the athlete has a short recovery period before the next bout of exercise (<12 h), a more aggressive approach is warranted: Athletes should consume approximately 1.5 L (51 ounces) of fluid (with sufficient electrolytes) for each kilogram of body weight loss (0.7 L or 24 oz or 3 cups for each pound of body weight). This amount of fluid helps account for increased urine production stemming from the consumption of a large volume of fluid (3, 147).

Fluid Intake Guidelines at a Glance

Before Training

- Athletes should have a USG reading <1.02 . They should prehydrate, if necessary, several hours before exercise to allow for fluid absorption and urine output (3).

During a Training Session

Children and Adolescents

- Children weighing 88 pounds (40 kg) should drink 5 ounces (148 mL) of cold water or a flavored, salted beverage every 20 minutes during training (34).
- Adolescents weighing 132 pounds (60 kg) should drink 9 ounces (266 mL) of cold water or a flavored, salted beverage every 20 minutes (171).

Adults

- Athletes should follow an individualized hydration plan. During prolonged activity in hot weather, they should consume a sport drink containing 20 to 30 mEq of sodium (460-690 mg with chloride as the anion) per liter, 2 to 5 mEq of potassium (78-195 mg) per liter, and carbohydrate in a concentration of 5% to 10%.

After Training

- Athletes should consume adequate food and fluids, as well as sodium, to restore hydration. If dehydration is significant or the athlete has <12 hours before the next exercise bout, a more aggressive approach is warranted and the athlete should consume approximately 1.5 liters (51 oz) of fluid (with sufficient electrolytes) for each kilogram of body weight lost (0.7 L or 24 oz or 3 cups for each pound of body weight).

Conclusion

Nutrition plays an important role in strength and conditioning. Adequate hydration and electrolytes; appropriate energy intake; and adequate protein, carbohydrate, fat, vitamin, and mineral intakes allow athletes to reap maximal benefits from training. A general understanding of nutrition principles and applications is essential for strength and conditioning professionals so that they can help athletes sort through nutrition misinformation and provide sound guidelines that athletes can use to improve their diets.

KEY TERMS

Acceptable Macronutrient Distribution Range (AMDR)

Adequate Intake (AI)

amino acids

anemia

bioavailability

carbohydrate

cholesterol

dehydration

Dietary Reference Intakes (DRIs)

disaccharides

docosahexaenoic acid (DHA)

eicosapentaenoic acid (EPA)

electrolytes

Estimated Average Requirement (EAR)

fat

fatty acids

fiber

fluid balance

fructose

galactose

gluconeogenesis

glucose
glycemic index (GI)
glycemic load
glycogen
glycogenesis
high-density lipoprotein (HDL)
hydration
hypohydration
hyponatremia
lactose
low-density lipoprotein (LDL)
macronutrient
maltose
micronutrient
minerals
monosaccharides
monounsaturated
muscle protein synthesis
MyPlate
nutrient density
omega-3 fatty acid alpha-linolenic acid (ALA)
polypeptide
polysaccharides
polyunsaturated
protein
protein digestibility
protein digestibility correct amino acid score (PDCAAS)
Recommended Dietary Allowances (RDA)
saturated
sports dietitian
sucrose
Tolerable Upper Intake Level (UL)
triglycerides
vegan

very low-density lipoproteins (VLDL)
vitamins

STUDY QUESTIONS

1. Which of the following is the player development staff member who has the most accurate nutrition knowledge and is best to counsel athletes with complex nutritional needs?
 - a. athletic trainer
 - b. strength and conditioning professional
 - c. sports dietitian
 - d. physical therapist
2. Which of the following food groups are illustrated in the MyPlate icon?
 - a. oils, proteins, water, fiber, vitamins, and minerals
 - b. protein foods, fruits, vegetables, grains, and dairy
 - c. proteins, carbohydrates, fats, and water
 - d. macronutrients, micronutrients, water
3. Which of the following is the recommended intake of protein for athletes?
 - a. 0.7 to 0.8 grams protein per kilogram body weight per day
 - b. 0.8 to 1.0 grams protein per kilogram body weight per day
 - c. 1.2 to 1.4 grams protein per kilogram body weight per day
 - d. 1.2 to 2.0 grams protein per kilogram body weight per day
4. For optimal health, which of the following is the recommended amount of added sugars in the diet?
 - a. less than 5% of total calories
 - b. less than 10% of total calories
 - c. more than 15% of total calories
 - d. more than 20% of total calories
6. Which of the following actions should athletes take to assess fluid losses after exercise?
 - a. check urine color
 - b. check their body temperature

- c. weigh themselves before and after exercise
- d. measure how much water they drank during exercise

TABLE 10.1 Food Group Recommendations From MyPlate

Population and age range		Estimated daily calories for those who are not physically fit ^a	Fruit ^b	Vegetables ^c	Grains ^d	Protein foods ^e	Dairy and fortified soy alternatives ^f
Children	2-3	1,000	1-1.5 cups	1 to 1.5 cups	3-5 oz equivalents (1.5-2.5 cups)	24 oz equivalents	2-2.5 cups
	4-8	1,200-1,400	1-2 cups	1.5-2.5 cups	4-6 oz equivalents (2-3 cups)	3-5.5 oz equivalents	2.5 cups
Girls	9-13	1,600	1.5-2 cups	1.5-3 cups	5-7 oz equivalents (2.5-3.5 cups)	4-6 oz equivalents	3 cups
	14-18	1,800	1.5-2 cups	2.5-3 cups	6-8 oz equivalents (3-4 cups)	5-6.5 oz equivalents	3 cups
Boys	9-13	1,800	1.5-2 cups	2-3.5 cups	5-9 oz equivalents (2.5-4.5 cups)	5-6.5 oz equivalents	3 cups
	14-18	2,200	2-2.5 cups	2.5-4 cups	6-10 oz equivalents (3-5 cups)	5.5-7 oz equivalents	3 cups
Females	19-30	2,000	1.5-2 cups	2.5-3 cups	6-8 oz equivalents (3-4 cups)	5-6.5 oz equivalents	3 cups

Population and age range		Estimated daily calories for those who are not physically fit ^a	Fruit ^b	Vegetables ^c	Grains ^d	Protein foods ^e	Dairy and fortified soy alternatives ^f
	31-59	1,800	1.5-2 cups	2-3 cups	5-7 oz equivalents (2.5-3.5 cups)	5-6 oz equivalents	3 cups
	60+	1,600	1.5-2 cups	2-3 cups	5-7 oz equivalents (2.5-3.5 cups)	5-6 oz equivalents	3 cups
Males	19-30	2,400	2-2.5 cups	3-4 cups	8-10 oz equivalents (4 to 5 cups)	6.5-7 oz equivalents	3 cups
	31-59	2,200	2-2.5 cups	3-4 cups	7-10 oz equivalents (3.5-5 cups)	6-7 oz equivalents	3 cups
	60+	2,000	2 cups	2.5-3.5 cups	6-9 oz equivalents (3-4.5 cups)	5.5-6.5 oz equivalents	3 cups

Population and age range	Estimated daily calories for those who are					Protein foods ^e	Dairy and fortified soy alternatives ^f
	not physically fit ^a	Fruit ^b	Vegetables ^c	Grains ^d			

The amount needed from each group depends on age, sex, height, weight, and level of physical activity. The amount can also depend on whether the individual is pregnant or breastfeeding. General recommendations by age are shown in the chart.

^aThese amounts are appropriate for individuals who get less than 30 minutes of moderate physical activity most days. Those who are more physically active may be able to consume more while staying within the daily caloric need. To find an individual's daily caloric need, go to www.myplate.gov/myplate-plan.

^bSee [table 10.2](#) for MyPlate Cup of Fruit Table.

^cSee [table 10.3](#) for MyPlate Cup of Vegetables Table.

^dSee [table 10.4](#) for MyPlate Ounce-Equivalent of Grains Table.

^eSee [table 10.5](#) for MyPlate Ounce-Equivalent of Proteins Table.

^fSee [table 10.6](#) for MyPlate Cup of Dairy Table.

Fruit. In general, 1 cup of fruit or 100% fruit juice or 1/2 cup of dried fruit can be considered as 1 cup from the Fruit Group.

Vegetables. Any vegetable or 100% vegetable juice counts as a member of the Vegetable Group. Vegetables may be raw or cooked; fresh, frozen, canned, or dried-dehydrated; and may be whole, cut up, or mashed. Vegetables in their nutrient-dense forms have limited additions such as salt, butter, or creamy sauces.

Grains. Any food made from wheat, rice, oats, cornmeal, barley, or another cereal grain is a grain product. Bread, pasta, breakfast cereals, tortillas, and grits are examples of grain products. Foods such as popcorn, rice, and oatmeal are included in the grains group. Half of one's grains should be whole grains. Whole grains have the entire kernel, which includes the bran, germ, and endosperm. Some whole grain examples are whole-wheat flour, bulgur, oatmeal, and brown rice.

Protein foods. All foods made from meat, poultry, seafood, beans and peas, eggs, processed soy products, nuts, and seeds are considered part of the Protein Foods Group. Beans and peas are also part of the Vegetable Group. In general, 1 ounce of meat, poultry, or fish; 1/4 cup cooked beans; 1 egg; 1 tablespoon of peanut butter; 1/2 ounce of nuts or seeds; 1/4 cup (about 2 oz) of tofu; or 1 ounce of tempeh, cooked, can be considered as 1 ounce-equivalent from the Protein Foods Group. Many Americans get the right amount of protein needed from meat, poultry, and eggs but do not meet the recommendations for seafood or nuts, seeds, and soy products. Meeting this can help increase the amount of important nutrients the body needs, like unsaturated fats, dietary fiber, and vitamin D.

Population and age range	Estimated daily calories for those who are not physically fit ^a	Fruit ^b	Vegetables ^c	Grains ^d	Protein foods ^e	Dairy and fortified soy alternatives ^f
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Dairy. The Dairy Group includes milk, yogurt, cheese, lactose-free milk and fortified soy milk, and yogurt. Foods made from milk that have little to no calcium and a high fat content, such as cream cheese, cream, and butter, are not considered dairy foods.

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TABLE 10.2 MyPlate Cup of Fruit Table

Fruit	Amount that counts as 1 cup of fruit
Apple	1 small or 1/2 large apple 1 cup, sliced or chopped, fresh 2/3 cup, baked 1/2 cup, dried
Applesauce	1 cup applesauce
Banana	1 large banana 1 cup, sliced 2/3 cup, mashed
Blueberries	1 cup, fresh or frozen 1/3 cup, dried
Cantaloupe	1 cup, diced or melon balls
Grapes	22 seedless grapes 1 cup, whole or cut up
Grapefruit	1 medium grapefruit 1 cup, sections
Kiwifruit	2 to 3 kiwifruits 1 cup, sliced or chopped
Mango	7 slices or chunks, fresh or frozen 1 cup, fresh or frozen 1/3 cup, dried

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Fruit	Amount that counts as 1 cup of fruit
Mixed fruit (fruit cocktail)	1 cup, diced or sliced, fresh or canned, drained
Orange	1 large orange 1 cup, sections
Peach	1 large peach 1 cup, sliced or diced, fresh, cooked, frozen or canned, drained 2 halves, canned
Pear	1 medium pear 1 cup, sliced or diced, fresh, cooked, or canned, drained
Pineapple	1 cup, chunks, sliced or crushed, fresh, cooked or canned, drained
Plum	3 medium or 2 large plums 1 cup, sliced, fresh or cooked 1/2 cup, dried (prunes)
Strawberries	About 8 large strawberries 1 cup, whole, halved, or sliced, fresh or frozen
Watermelon	1 small wedge or slice 1 cup, diced or melon balls
Dried fruit (e.g., raisins, prunes, apricots, figs)	1/2 cup dried fruit
100% fruit juice (e.g., orange, apple, grape, grapefruit)	1 cup

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TABLE 10.3 MyPlate Cup of Vegetable Table

Vegetable group	Vegetable	Amount that counts as 1 cup of vegetables
Dark green vegetables	Broccoli	1 cup, chopped or florets, fresh or frozen

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Vegetable group	Vegetable	Amount that counts as 1 cup of vegetables
	Amaranth leaves, beet greens, Bok choy, broccoli rabe (rapini), chard, collards (collard greens), cress, dandelion greens, kale, mustard greens, spinach, Swiss chard, watercress	1 cup, cooked 2 cups, fresh
	<i>Raw leafy greens:</i> arugula (rocket), basil, cilantro, dark green leafy lettuce, endive, escarole, mixed greens, mesclun, romaine	2 cups, fresh
Red and orange vegetables	Carrots	2 medium carrots 1 cup, sliced or chopped, fresh, cooked or frozen 1 cup baby carrots
	Pumpkin, calabaza	1 cup, mashed, cooked
	Red and orange bell peppers	1 large bell pepper 1 cup, chopped, fresh, or cooked
	Sweet potato	1 large sweet potato, baked 1 cup, sliced or mashed, cooked
	Tomatoes	1 large tomato 2 small tomatoes 1 cup, chopped or sliced, fresh, canned, or cooked
	100% vegetable juice	1 cup
	Winter squash (acorn, butternut, Hubbard, kabocha)	1 cup, cubed, cooked
Beans, peas, and lentils	Dry beans and peas and lentils (bayo, black, brown, fava, garbanzo, kidney, lima, mung, navy, pigeon, pink, pinto, soy, or white beans; black-eyed peas	1 cup, whole or mashed, cooked

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Vegetable group	Vegetable	Amount that counts as 1 cup of vegetables
	[cow peas] or split peas; and red, brown, and green lentils)	
Starchy vegetables	Cassava	3/4 cup, cooked
	Corn, yellow or white	1 large ear of corn 1 cup corn kernels, fresh or frozen
	Green peas	1 cup, fresh or frozen
	Plantains	3/4 cup, cooked
	White potatoes	1 medium white potato, boiled or baked 1 cup, diced, mashed, fresh or frozen
Other vegetables	Asparagus	1 cup
	Avocado	1 avocado
	Bean sprouts	1 cup, cooked
	Cabbage, green, red, napa, savoy	1 cup, chopped or shredded raw or cooked
	Cauliflower	1 cup, pieces or florets raw or cooked, fresh or frozen
	Celery	1 cup, diced or sliced, raw or cooked 2 large stalks (11" to 12" long)
	Cucumbers	1 cup, raw, sliced or chopped

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Vegetable group	Vegetable	Amount that counts as 1 cup of vegetables
	Green or wax beans	1 cup, raw or cooked
	Green bell peppers	1 large bell pepper 1 cup, chopped, raw or cooked, fresh or frozen
	Lettuce, iceberg or head	2 cups, raw, shredded or chopped
	Mushrooms	1 cup, raw or cooked
	Okra	1 cup, cooked
	Onions	1 cup, chopped, raw or cooked
	Summer squash or zucchini	1 cup, cooked, sliced or diced

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TABLE 10.4 MyPlate Ounce-Equivalent of Grains Table

Grain	Amount that counts as 1 oz equivalent of grains
Bagels	2" mini bagel
Bagel or pita chips	1/3 cup bagel or pita chips
Barley	1/2 cup, cooked
Buckwheat	1/2 cup, cooked
Breads	1 regular slice of bread 1 small slice of French bread 4 snack-size slices of rye bread

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Grain	Amount that counts as 1 oz equivalent of grains
Bulgur	1/2 cup, cooked
Challah bread	1 medium or regular sliced
Cornbread	1 small piece
Couscous	1/2 cup, cooked
Crackers	5 whole wheat crackers 2 rye crisp breads 7 square or round crackers
English muffins	1/2 English muffin
Muffins	1 small muffin
Oatmeal	1/2 cup, cooked 1 packet instant 1 oz (1/3 cup) dry (regular or quick)
Pancakes	1 pancake (4-1/2" diameter) 2 small pancakes (3" diameter)
Pasta (spaghetti, macaroni, noodles)	1/2 cup, cooked 1 oz, dry
Popcorn	3 cups, popped
Quinoa	1/2 cup, cooked
Ready-to-eat breakfast cereal	1 cup, flakes or rounds 1-1/4 cup, puffed
Rice	1/2 cup, cooked 1 oz, dry
Tortillas	1 small flour tortilla (6" diameter) 1 corn tortilla (6" diameter)

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TABLE 10.5 MyPlate Ounce-Equivalent of Proteins Table

Protein	Amount that counts as 1 oz equivalent of protein
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Protein	Amount that counts as 1 oz equivalent of protein
Meats	1 oz cooked lean beef, goat, ham, lamb, or pork 1 oz cooked lean ground beef or pork 1 slice of luncheon or deli meats (beef, chicken, ham, pork, turkey) 1 oz cooked game meats (bear, bison, deer, elk, moose, opossum, rabbit, venison) 1 oz cooked organ meats
Poultry	1 oz cooked (without skin) chicken, ostrich, or turkey 2 oz cooked Cornish hen, duck, goose, pheasant, or quail 1 sandwich slice of turkey or chicken breast (4-1/2" × 2-1/2" × 1/8")
Seafood	1 oz cooked finfish (black sea bass, catfish, cod, flounder, freshwater trout, haddock, hake, halibut, herring, light tuna, mackerel, mullet, perch, pollock, salmon, sea bass, snapper, sole, tilapia, whiting) 1 oz cooked shellfish (clams, crab, crayfish, lobster, mussels, octopus, oysters, scallops, shrimp, squid [calamari]) 1 oz canned fish (anchovies, freshwater trout, herring, light tuna, salmon, sardines)
Eggs	1 egg 1-1/2 egg whites (or 3 tbsp liquid egg white product)
Nuts, seeds, and soy products	1/2 oz of nuts (12 almonds, 24 pistachios, 7 walnut halves) 1/2 oz of seeds (chia, flax, pumpkin, sesame, sunflower, or squash seeds), hulled, roasted 1 tbsp of almond, cashew, peanut, or sunflower butter, or sesame paste (tahini) 1/4 cup (about 2 oz) of tofu 1 oz tempeh, cooked
Beans, peas, and lentils	1/4 cup of cooked beans, peas, or lentils (bayo, black, brown, fava, garbanzo, kidney, lima, mung, navy, pigeon, pink, pinto, soy, or white beans; black-eyed peas [cow peas] or split peas; and red, brown, and green lentils) 1/4 cup of baked beans or refried beans 1 falafel patty (2-1/4", 4 oz) 6 tbsp hummus

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TABLE 10.6 MyPlate Cup of Dairy Table

Dairy	Amount that counts as 1 cup of dairy
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Dairy	Amount that counts as 1 cup of dairy
Milk	1 cup milk 1 half-pint container milk 1/2 cup evaporated milk 1 cup calcium-fortified soy milk 1 half-pint container calcium-fortified soy milk
Yogurt	1 cup yogurt (dairy or fortified soy)
Cheese	1-1/2 oz hard cheese (cheddar, mozzarella, Swiss, Parmesan) 1/3 cup shredded cheese 1 oz processed cheese (American) 1/2 cup ricotta cheese 2 cups cottage cheese 2 oz queso fresco 2 slices queso blanco

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